Module 5 Activity 1: Working with Rasters

Purpose:

Incorporate the ***arcpy.sa*** module to work with geoprocessing tools to create and analyze raster data. List and describe the resulting raster datasets.

Instructions:

Any Python environment will work for this exercise.

Download the data Module 5 Activity 1 and save the data to your working directory.

1. Import the ***arcpy, os*** modules
2. Import the ***arcpy.sa*** module
3. Check out the spatial analyst extension so it can be used
4. Add a ***print*** statement to the ***CheckOutExtension*** function to notify the user that it is ready for use
5. Set the workspace and overwrite outputs

Python

```
#import arcpy, os and Spatial extension
import arcpy, os
from arcpy import env
from arcpy.sa import *
#check out the spatial extension
arcpy.CheckOutExtension("Spatial")
print ("Spatial extension is ready to use.")
#set the overwrites and workspace for the shapefiles
env.overwriteOutput = True
env.workspace = r"C:\GEOS456"
```

The statement ***from arcpy.sa import **** imports all the functions used in the ***arcpy.sa*** module. As mentioned in the slides, this will allow you to call the tools directly rather than referencing the ***arcpy.sa*** module for every tool used.

6. Set the workspaces for the geodatabase and the feature dataset to store the converted shapefiles and the where the rasters will be saved

```
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from arcpy.sa import *
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print ("Spatial extension is ready to use.")
#set the overwrites and workspace for the shapefiles
env.overwriteOutput = True
env.workspace = r"C:\GEOS456"
#set the workspace for the geodatabases for the converted .shp and where the rasters will be saved
gdb = r"C:\GEOS456\RasterData.gdb"
gdb_FD = r"C:\GEOS456\RasterData.gdb\BaseData"
```

7. Use the ***os*** module **strip** method to “strip away” the .shp extension
8. Use the ***CopyFeatures_management*** function to copy the shapefiles to the ***RasterData.gdb*** provided.

```
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gdb_FD = r"C:\GEOS456\RasterData.gdb\BaseData"
#copy all the shapefiles to the feature dataset
shpList = arcpy.ListFeatureClasses()
for shp in shpList:
    outFC = os.path.join(gdb_FD, shp.strip(".shp"))
    arcpy.CopyFeatures_management(shp, outFC)
```

The next steps will use the ***Interpolation*** toolset in the spatial analyst toolbox to create 3 different surfaces from the ***Temperature*** feature class. Make sure to review the syntax for ***IDW, Krig, Spline***.

9. Define the point feature class which will serve as the input for the raster datasets
10. Generate the raster surfaces

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#copy all the shapefiles to the feature dataset
shpList = arcpy.ListFeatureClasses()
for shp in shpList:
    outFC = os.path.join(gdb_FD, shp.strip(".shp"))
    arcpy.CopyFeatures_management(shp, outFC)
#define the point feature class for the interpolation
point = gdb_FD + "\\" + "Temperature"
#generate the raster surfaces
outIDW = Idw(point, "TEMP")
outKrig = Kriging(point, "TEMP", "SPHERICAL")
outSpline = Spline(point, "TEMP")
```

11. Use the ***save*** method to permanently save the resulting rasters to the geodatabase provided

Raster datasets cannot be saved inside a feature dataset and it is not necessary to save the rasters inside a raster catalog for the purposes of this activity.

```
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shpList = arcpy.ListFeatureClasses()
for shp in shpList:
    outFC = os.path.join(gdb_FD, shp.strip(".shp"))
    arcpy.CopyFeatures_management(shp, outFC)
#define the point feature class for the interpolation
point = gdb_FD + "\\" + "Temperature"
#generate the raster surfaces
outIDW = Idw(point, "TEMP")
outKrig = Kriging(point, "TEMP", "SPHERICAL")
outSpline = Spline(point, "TEMP")
#save the rasters to the gdb
outIDW.save(gdb + "\\" + "Temp_IDW")
outKrig.save(gdb + "\\" + "Temp_Krig")
outSpline.save(gdb + "\\" + "Temp_Spline")
```

At this point in the script, the rasters have been generated and could be accessed directly from the geodatabase. The next steps will gather a list of the raster datasets generated as well as a description of each raster using the ***ListRasters*** function and the ***Describe*** function.

12. Make sure the workspace is pointing to the geodatabase where the rasters are saved
13. Create a variable to hold the ***ListRasters*** function
14. Start a ***for*** loop to iterate through the geodatabase
15. Within the ***for*** loop, describe the raster format, cell size and spatial reference for each raster surface
16. Use a ***print*** statement to return a full list of the rasters generated

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    arcpy.CopyFeatures_management(shp, outFC)
#define the point feature class for the interpolation
point = gdb_FD + "\\" + "Temperature"
#generate the raster surfaces
outIDW = Idw(point, "TEMP")
outKrig = Kriging(point, "TEMP", "SPHERICAL")
outSpline = Spline(point, "TEMP")
#save the rasters to the gdb
outIDW.save(gdb + "\\" + "Temp_IDW")
outKrig.save(gdb + "\\" + "Temp_Krig")
outSpline.save(gdb + "\\" + "Temp_Spline")
#get a list of the rasters in the gdb
env.workspace = gdb
rasterList = arcpy.ListRasters()
for rasters in rasterList:
    #describe some raster properties
    rasterDesc = arcpy.Describe(rasters)
    print (rasters)
    print ("Raster Format:", rasterDesc.format)
    print ("Cell Size:", rasterDesc.meanCellWidth)
    print ("Spatial Reference:", rasterDesc.spatialReference.name)
    print ("")
```

17. Finally, check the spatial analyst extension back in to ensure it can be used again by someone else
18. Use a ***print*** statement to inform the user that the license has been checked back in and returned to the license manager

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outIDW = Idw(point, "TEMP")
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#save the rasters to the gdb
outIDW.save(gdb + "\\" + "Temp_IDW")
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for rasters in rasterList:
    #describe some raster properties
    rasterDesc = arcpy.Describe(rasters)
    print (rasters)
    print ("Raster Format:", rasterDesc.format)
    print ("Cell Size:", rasterDesc.meanCellWidth)
    print ("Spatial Reference:", rasterDesc.spatialReference.name)
    print ("")
#check the spatial extension back in
arcpy.CheckInExtension("Spatial")
print ("Spatial extension checked back in.")
```

19. Run the script
20. Examine your results

Python

```
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outSpline = Spline(point, "TEMP")
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for rasters in rasterList:
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    rasterDesc = arcpy.Describe(rasters)
    print (rasters)
    print ("Raster Format:", rasterDesc.format)
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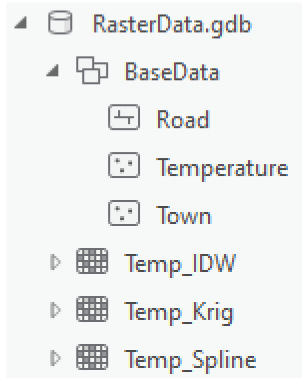
    print ("Spatial Reference:", rasterDesc.spatialReference.name)
    print ("")
#check the spatial extension back in
arcpy.CheckInExtension("Spatial")
print ("Spatial extension checked back in.")
Spatial extension is ready to use.
Temp_IDW
Raster Format: FGDBR
Cell Size: 441.8028467999995
Spatial Reference: NAD_1983_UTM_Zone_11N

Temp_Krig
Raster Format: FGDBR
Cell Size: 441.8028467999995
Spatial Reference: NAD_1983_UTM_Zone_11N

Temp_Spline
Raster Format: FGDBR
Cell Size: 441.8028467999995
Spatial Reference: NAD_1983_UTM_Zone_11N

Spatial extension checked back in.
```

21. Check the geodatabase in ArcCatalog to ensure everything is there



22. Add the newly created raster datasets to Pro to view your results and compare the resulting raster surfaces

<u>NOTE:</u> remember that raster datasets need to be kept short. Errors can occur in the generation of rasters that exceed 13 characters. In this activity, we used the *Temperature* feature class to generate rasters and the resulting output name *Temp_IDW*. Use caution when using the name *Temp* as it can be thought of as a *<u>temporary</u>* file. Ensure proper data management techniques to avoid confusion.

<u>Next steps:</u>

1. Use different raster descriptions in the ***Describe*** to return more information about the raster datasets created in this activity.